```
#include <stdio.h>
#include <string.h>

#define MAX 5

struct Job
{
   int id;
   char title[50];
};

struct Job queue[MAX];

int front = -1;
int rear = -1;

// Add print job
void addJob(){
   int id;
   char title[50];

   if ((rear + 1) % MAX == front)
   {
      printf("Queue Overflow! Print queue is full.\n");
      return;
   }

   printf("Enter Job ID: ");
   scanf("%d", &id);

   printf("Enter Document Title: ");
   scanf("%49s", title);

   if (front == -1)
      front = 0;

   rear = (rear + 1) % MAX;

   queue[rear].id = id;
   strcpy(queue[rear].title, title);

   printf("Print job added successfully.\n");
}
```

```
// Process print job
void processJob(){
   if (front == -1)
   {
      printf("Queue Underflow! No print jobs available.\n");
      return;
   }

   printf("Processing Job ID: %d\n", queue[front].id);
   printf("Document: %s\n", queue[front].title);

   if (front == rear)
   {
      front = -1;
      rear = -1;
   }
   else
   {
      front = (front + 1) % MAX;
   }
   printf("Print job processed successfully.\n");
}

// Display all pending jobs
void displayJobs()
{
   int i;

   if (front == -1)
   {
      printf("No pending print jobs.\n");
      return;
   }

   printf("\nPending Print Jobs:\n");

   i = front;

   while (1)
   {
      printf("Job ID: %d, Document: %s\n",
           queue[i].id, queue[i].title);

      if (i == rear)
         break;
```

```
    i = (i + 1) % MAX;
  }
}

// Main function
int main(){
  int choice;

  do
  {
    printf("\n--- PRINTER QUEUE MANAGER ---\n");
    printf("1. Add Print Job\n");
    printf("2. Process Print Job\n");
    printf("3. Display Pending Jobs\n");
    printf("0. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
      case 1:
        addJob();
        break;

      case 2:
        processJob();
        break;

      case 3:
        displayJobs();
        break;

      case 0:
        printf("Exiting program.\n");
        break;

      default:
        printf("Invalid choice!\n");
    }
  } while (choice != 0);

  return 0;
}
```